

Title

Comprehensive geriatric rehabilitation model reduces functional decline in acute hospital settings: outcomes from a cluster-randomized stepped-wedge implementation trial

Abstract

In older adults, hospitalization often precipitates substantial loss of mobility and independent functioning. We tested whether an integrated care package combining structured mobility rehabilitation, cognitive support protocols, and simplified medication management could preserve independence in elderly inpatients across departments of a university hospital. This stepped-wedge cluster-randomized trial enrolled 589 older patients (mean age 79.6 years, 48% female) stratified by hospital ward. The intervention integrated supervised physical rehabilitation sessions, delirium prevention protocols, simplified medication regimens, and structured care coordination. Physical capacity and functional independence were assessed at hospital admission, discharge, and 6-month follow-up. Patients assigned to the intervention demonstrated significantly greater improvements in physical capacity from admission to discharge, with sustained benefits through 6-month follow-up. Functional independence scores improved substantially in the intervention group. These findings reinforce the value of early, systematic mobility and supportive-care strategies in counteracting hospitalization-related functional decline in older populations.